

---

## CE 509: Bridge Engineering

**Course Code: CE 509**

**Course Name: Bridge Engineering**

L-T-P-C: 3-0-0-3

Prerequisite: CE 301 and CE 351

Students intended for: B.Tech. 3rd/4th year/PG

Elective or Core: Elective

Approval: 15th Senate

### Course contents

- **Introduction**

Importance of bridges; Historical development and classification of bridge types; Various structural forms; Materials in bridge construction; Consideration for site selection; Determination of design discharge; linear waterways and economical span; High flood level (HFL) and vertical clearance; Traffic projection. [4 Lectures]

- **Design Loads**

Consideration of various forces and their application as per IRC specifications for loadings on road bridges. [3 Lectures]

- **Structural Behavior and Concepts for Analysis**

Moving load analysis; Grillage analogy; Structural behavior of box-girder bridges, arch bridges, suspension bridges, and cable stayed bridges. [8 Lectures]

- **Design of Bridge Superstructure**

Analysis and design of reinforced concrete slab bridge deck; Design considerations for skewed slab decks; Pretensioning and post-tensioning; Analysis and design of pre-stressed concrete slab deck; Introduction to T-beam bridges; Approximate methods for analysis of T-beam bridges: Courbon's method; Illustrative examples. [17 Lectures]

- **Bridge Bearings and Joints**

Importance and types of bearings; Design of elastomeric bearings: Seismic considerations for bearing design; Provisions for expansion joints. [5 Lectures]

- **Bridge Substructure**

Components of substructure; Computation of different forces acting on substructure; Introduction to various bridge foundation types. [3 Lectures]

- **Construction and Maintenance**

Common bridge construction techniques; overview of segmental construction for long span bridges; Primary [2 Lectures]

---

### Text Books

1. D.J. Victor, **Essentials of Bridge Engineering**, 6th Edition, Oxford and ffH Publ ishing, New Delhi, India, 2007
2. N.K. Raju, **Design of Bridges**, 4th Edition, Oxford and TBH Publi shing, New Delhi, 2009.

### Reference Books:

1. E.C. Hambly, **Bridge Deck Behaviour**, 2nd Edition, CRC Press, Taylor and Francis Group, New York, USA, 1991.
2. IRC-112 (20 II), Code of Practice for Concrete Road Bridges , Indian Roads Congress, New Delhi, India.
3. IRC-6 (20 I 4), Standard Specifications and Code of Practice for Road Bridges, Section: II - Loads and Stresses , Indian Roads Congress, NeDelhi , India.
4. N. Rajagopalan, **Bridge Superstructure**, Narosa Publi shing House, New Delhi, India, 2013.
5. S. Ponnuswamy, **Bridge Engineering**, 2nd Edition, Tata McGraw-Hill Education, New Delhi, India, 2007.
6. V.K. Raina, **Rainas Concrete Bridge Practice Analysis, Design & Economics**, 4th Edition, Shroff Pub-lishers and Distributors Pvt. Ltd., Mumbai, India, 2014.